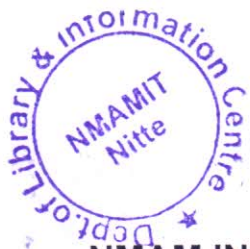


EC



USN

--	--	--	--	--	--	--	--	--	--

NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

Seventh Semester B.Tech (CBCS) Degree Examinations

December 2025

EC3001-1 – ANTENNA AND MICROWAVE SYSTEMS

(For ECE)

Duration: 3 Hours

Max. Marks: 100

Note: Usage of SMITH CHART is Permitted.

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five** full questions choosing **Two** full questions from **Unit – I & Unit – II** each and **One** full question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- In a microstrip line, the fields are predominantly
 - TEM
 - Quasi-TEM
 - TE only
 - TM only
- A $\lambda/4$ short-circuited stub behaves as
 - Open circuit
 - Short circuit
 - Inductor
 - Conductor
- Reflex klystron frequency is tuned mainly by
 - Repeller voltage
 - Magnetic field
 - Collector bias
 - Heater Current
- A Magic-Tee is also called:
 - Hybrid Tee
 - Directional coupler
 - E-plane Tee only
 - H-plane Tee only
- For wide microstrip lines, the characteristic impedance:
 - Increases
 - Decreases
 - Remains constant
 - Becomes infinite
- On a lossless Smith chart, moving toward the load corresponds to rotation
 - Clockwise
 - Counter-clockwise
 - Radial inward
 - Radial outward
- In the Smith chart, a constant- $|\Gamma|$ circle corresponds to constant
 - Resistance
 - Reactance
 - VSWR
 - Conductance
- A quarter-wave transformer repeats the load impedance every
 - $\lambda/2$
 - λ
 - $\lambda/4$
 - $3\lambda/4$
- Which parameter defines the ability of an antenna to concentrate energy in a particular direction?
 - Gain
 - Directivity
 - Efficiency
 - Radiation Resistance
- For an isotropic radiator, directivity is:
 - 0 dB
 - 1 dB
 - 2 dB
 - Infinite
- An antenna with half-power beamwidth (HPBW) of 60° has approximate directivity:
 - 3 dB
 - 6 dB
 - 12 dB
 - 30 dB

12. Radiation intensity is maximum in the direction of:
 - A) Null
 - B) Side lobe
 - C) Main lobe
 - D) End-fire
13. For an array of isotropic radiators, maximum directivity is:
 - A) Proportional to N
 - B) Proportional to N^2
 - C) Inversely proportional to N
 - D) Independent of N
14. The first null beamwidth (FNBW) is approximately:
 - A) $2 \times \text{HPBW}$
 - B) Equal to HPBW
 - C) Smaller than HPBW
 - D) Independent of HPBW
15. The bandwidth of Yagi is:
 - A) Narrow
 - B) Very wide
 - C) Infinite
 - D) None
16. By Babinet's principle, fields of a slot are related to:
 - A) Fields of complementary dipole
 - B) Horn radiation
 - C) Reflector
 - D) Waveguide
17. Slot antennas are often cut in:
 - A) Waveguides
 - B) Parabolic reflectors
 - C) Dipoles
 - D) Coaxial lines
18. Horn antenna is an extension of
 - A) Dipole
 - B) Waveguide
 - C) Loop
 - D) Reflector
19. Reflector antennas are widely used in:
 - A) Satellite communication
 - B) AM radio
 - C) TV indoor reception
 - D) Handheld radios
20. Rumsey's principle states:
 - A) Antenna shape defined by angles $\rightarrow$ frequency independent
 - B) Antenna defined by length only
 - C) Antenna defined by dielectric
 - D) Antenna defined by waveguide cutoff

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) With a neat construction and applegate diagram, explain the principle of operation of a Reflex klystron.	8	L*2	1	1
	b) What is S-matrix? Write the properties of S- matrix.	8	L2	1	1
2.	a) With a neat diagram of a transmission line, derive the necessary conditions required for matching a line.	8	L2	2	1
	b) A microstrip line uses a substrate with $\epsilon_r = 9.7$ (alumina) and $d = 0.5$ mm. The strip width is $w = 0.5$ mm. Find E_{refr} Z_o and the microstrip wavelength at $f = 2$ GHz. consider $\tan \delta = 0.2$ mm. if $f = 4$ GHz, calculate the total attenuation (assume metal as copper).	8	L3	2	1
3.	a) With a neat construction diagram explain a microstripline. Write its characteristics.	8	L2	2	1
	b) Explain the working of a E-H Plane Tee Junction with neat diagrams, and derive S – Matrix.	8	L2	2	1
Unit – II					
4.	a) Explain the following terms in connection with antenna parameters				
	(i) Beam area				
	(ii) Radiation intensity				
	(iii) Beam efficiency				
	(iv) Stray factor				
	Write the expression in each case.	8	L2	3	1

b)	Define directivity. Derive an expression for directivity and hence deduce a relationship with effective aperture.	8	L2	3	1
5. a)	Derive the Relation of the Poynting vector S and the two electric field components of the far field.	8	L2	4	1
b)	Derive and plot the radiation pattern for an array of two isotropic sources of equal amplitude and opposite phase, calculate HPBW.	8	L2	4	1
6. a)	Derive array factor for 'n' isotropic point sources of equal amplitude and spacing, show the vector addition of the field with 5 isotropic sources.	8	L3	4	1
b)	Derive the equation for scalar and vector potential for short dipole , and give the equation for scalar potential.	8	L2	4	1
Unit – III					
7. a)	Explain the concept of Babinet's principle and Complementary antenna.	8	L2	5	1
b)	Design a 6 element typical modern-version 6-element Yagi-Uda antenna (folded $\lambda/2$ dipole) fed by a 300- Ω , 2-wire transmission line working at 45 MHz (twin-line), a reflector and 4 directors.	8	L3	5	1
8. a)	Explain the constructional and Monofilar axial-mode helical antenna.	8	L2	5	1
b)	Explain the construction and operating principles of log- Periodic dipole array antenna.	8	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
